

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	15-07-2026
Team ID	
Project Name	ShopEZ – Full Stack MERN E-Commerce Platform
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Register using email and password
FR-2	User Confirmation	Confirmation via Email Confirmation via OTP
FR-3	User Authentication	Login, Logout, JWT Authentication
FR-4	Product Management	View, Search, Filter Products
FR-5	Shopping Cart	Add, Update, Remove Cart Items
FR-6	Wishlist	Add and Remove Wishlist Items

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Easy-to-use and responsive interface
NFR-2	Security	JWT Authentication, bcrypt password hashing, protected routes
NFR-3	Reliability	Stable API responses and error handling
NFR-4	Performance	Fast page loading and optimized database queries
NFR-5	Availability	Accessible 24×7
NFR-6	Scalability	MVC architecture and modular codebase